EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are:
   a. 5, 1, 2 b. 5, 1, 5
   c. 5, 5, 2 d. 4, 4, 2
2. Which among the following is the supplementary Quantity:
   a. Mass b. Time
   c. Solid angle d. Luminosity
3. The unit of solid angle is
   a. Second b. Steradian
   c. Kilogram d. Candela
4. The quantity having the same unit in all system of unit is:
   a. mass b. time
   c. length d. temperature
5. Random errors can be eliminated by:
   a. taking number of observations and their mean.
   b. measuring the quantity with more than one instrument
   c. eliminating the cause
   d. careful observations
6. Systemic error can be:
   a. either positive or negative c. positive only
   b. negative only d. zero error
7. $[MLT^{-2}]$ is dimensional formula of:
   a. strain c. displacement
   b. force d. pressure
8. Which of the following pair has the same dimension?
   a. moment of inertia and torque c. impulse and momentum
   b. surface tension and force d. specific heat and latent heat
9. Dependent variable is:
   a. cause c. cause and effect
   b. effect d. reason
10. Dimensions of kinetic energy is the same as that of:
    a. Acceleration c. Work
    b. Velocity d. Force

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. To get a resultant displacement of 10m, two displacement vectors of magnitude 6m and 8m should be combined:
   (a) Parallel (b) Antiparallel
   (c) At an angle of 45^0 (d) Perpendicular to each other
2. The velocity of a particle at an instant is 10 m/s and after 5sec the velocity of particle is 20m/s. The velocity at 3 sec before in before 10m/s is:
   (a) 8 (b) 4
   (c) 6 (d) 7
3. A ball is thrown upwards with a velocity of 100 m/s. It will reach the ground after:
   (a) 10 sec (b) 20 sec
   (c) 5 sec (d) 40 sec
4. Two projectiles are fired from the same point with the same speed at angles of projection 60^0 and 30^0 respectively. Which one of the following is true?
   (a) The range will be same (b) Their maximum height will be same
   (c) Their landing velocity will be same (d) their time of flight will be same
5. The ratio of numerical values of average velocity and average speed of a body is always:
   (a) Unity (b) Unity or less
   (c) Unity or more (d) Less than unity
6. If the average velocities of a body become equal to the instantaneous velocity, body is said to be moving with:
   (a) Uniform acceleration (b) Uniform velocity
   (c) Variable velocity (d) Variable acceleration
7. At the top of a trajectory of a projectile, the acceleration is:
   (a) maximum (b) minimum
   (c) zero (d) g
8. At what angle the range of projectile becomes equal to the height of projectile?
   (a) 65^0 (b) 45
   (c) 76^0 (d) 30^0
9. The angle at which dot product becomes equal to the cross product is:
   (a) 65^0 (b) 45^0
   (c) 76^0 (d) 30^0
10. If the dot product of two non-zero vectors vanishes; the vectors will be:
    (a) in the same direction (b) opposite direction to each other
    (c) perpendicular to each other (d) zero



- Dynamics: Study of the forces and torques that cause motion and the resulting effects.
- Force: Push or pull on an object; causes acceleration.
- Newton's Laws of Motion:
- First Law (Law of Inertia): An object at rest stays at rest, and an object in motion stays in motion unless acted upon by an external force.
- Second Law (F = ma): The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.
- Third Law (Action-Reaction): For every action, there is an equal and opposite reaction.
- Inertia: Resistance of an object to changes its state in its motion due to its mass.
- Friction: Resistance force between two surfaces in contact; opposes relative motion.
- Normal Force: Force exerted by a surface to support the weight of an object in contact with it.
- Momentum: Product of an object's mass and velocity or quantity of motion; conserved in a closed system.
- Impulse: Change in momentum of an object due to an applied force over a time interval.
- Conservation of Momentum: In a closed system, the total momentum before a collision is equal to the total momentum after the collision.
- Elastic Collision: Collisions where both momentum and kinetic energy are conserved.
- Inelastic Collision: Collisions where momentum is conserved, but kinetic energy is not.



Section (A): Multiple Choice Questions (MCQs)

1. The rate of change of linear momentum of a body is called a:
   a) Linear force b) Angular force
   c) Power d) Impulse
2. The term mass refers to the same physical concept as:
   a) weight b) inertia
   c) force d) acceleration
3. Which one of the following force is also called as self adjusting force ?
   a) frictional force b) tension
   c) Weight d) Thrust
4. The laws of motion shows the relationship between:
   a) velocity and acceleration b) mass and velocity
   c) mass and acceleration d) force and acceleration

5. The motion of rocket in the space is according to the law of conservation of:
   a) Energy b) linear momentum
   c) mass d) angular momentum
6. A bomb of mass 12 kg initially at rest explodes into two pieces of masses 4kg and 8kg . The speed of 8kg mass is 6 m/s. The kinetic energy The kinetic energy of 4 kg mass is:
   a) 32 J b) 48 J
   c) 114 J d) 288 J
7. If momentum is increased by 20% then K.E increases by:
   a) 44% b) 55%
   c) 66% d) 77%
8. The kinetic energy of body of mass 2kg and momentum of 2 Ns is:
   a) 1 J b) 2 J
   c) 3 J d) 4 J
9. For the same kinetic energy, the momentum is maximum for:
   a) an electron b) a proton
   c) a deuteron d) alpha particle
10. A 3kg bowling ball experiences a net force of 15N. What will be its acceleration.
   a) 35 m/s2 b) 7 m/s2
   c) 5 m/s2 d) 35 m/s2

Section (B): Structured Questions

CRQs:

1. State Newton's second law of motion.
2. How does mass affect an object's acceleration?
3. What are the different types of forces that can act on an object?
4. How do you calculate net force?
5. State the law of conservation of momentum.
6. What is the difference between elastic and inelastic collisions?
7. How does impulse relate to force and time?
8. How does friction influence the motion of an object?

ERQs:

1. Explain Newton's Second Law of Motion and how it relates force, mass, and acceleration. Provide an example to illustrate the concept.
2. How does the conservation of momentum apply in each type of collision? Give real-life examples of both types of collisions.
3. Explain how the concept of impulse is related to the change in momentum of an object. Provide an example of an everyday life where impulse plays a significant role.
4. State the law of conservation of momentum. Provide an example from everyday life that demonstrates the principle of momentum conservation.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. One radian is about
   a) 25°
   b) 37°
   c) 45°
   d) 57°
2. Wheel turns with constant angular speed then:
   a) each point on its rim moves with constant velocity
   b) each point on its rim moves with constant acceleration
   c) the wheel turns through equal angles in equal times
   d) the angle through which the wheel turns in each second increases as time goes on
   e) the angle through which the wheel turns in each second decreases as time goes on
3. The rotational inertia of a wheel about its axle does not depend upon its
   a) diameter
   b) mass
   c) distribution of mass
   d) speed of rotation
4. A force with at given magnitude is to be applied to a wheel. The torque can be maximized by
   a) applying the force near the axle, radially outward from the axle.
   b) applying the force near the rim, radially outward.
   c) applying the force near the axle, parallel to a tangent to the wheel.
   d) applying the force at the rim, tangent to the rim
5. An object rotating about a fixed axis, I is its rotational inertia and α is its angular acceleration. Its
   a) is the definition of torque
   b) is the definition of rotational inertia
   c) is definition of angular acceleration
   d) follows directly from Newton's second law.
6. The angular momentum vector of Earth about its rotation axis, due to its daily rotation is, directed
   a) tangent to the equator towards east
   b) tangent to the equator towards the west
   c) north
   d) towards the Sun.
7. A stone of 2 kg is tied to a 0.50 m long string and swung around a circle at angular velocity of 12 rad/s . The net torque on the stone about the center of the circle is
   a) 0 N.m
   b) 6 N.m
   c) 12 N.m
   d) 72 N.m
8. A man, with his arms at his sides, is spinning on a light frictionless turntable . When he extends his arms
   a) his angular velocity increases
   b) his angular velocity remains same
   c) his rotational inertia decreases
   d) his angular momentum remains the same.

9. A space station revolves around the earth as a satellite, 100 km above the Earth's surface. What is the net force on an astronaut at rest inside the space station?
   a) equal to her weight on earth
   b) a little less than her weight on earth
   c) less than half her weight on earth
   d) zero (she is weightless)
10. If the external torque acting on a body is zero, then its
   a) angular momentum is zero
   b) angular momentum is conserved
   c) angular acceleration is maximum
   d) rotational motion is maximum

Section (B): Structured Questions

CRQ:

1. For an isolated rotating body, what is the relation between angular velocity and radius?
2. When the moment of inertia of a rotating body is halved, then what will be the effect on angular velocity?
3. Compare kinematics equation of linear motion and circular motion
4. Can a small force ever exert a greater torque than a larger force? Give reason.
5. Give two real world applications of angular momentum
6. Derive relationship between torque and angular acceleration.
7. List the moment of inertia dependent factors.

ERQ

1. State and explain the law of conservation of angular momentum. Give two examples to illustrate it.
2. Discuss forces action on banked curve and derive relation between curve angle and velocity?
3. Derive the formula for centripetal acceleration using fundamental principles and equations, illustrating each step of the derivation
4. Derive the formula for the moment of inertia of a uniform rod or a solid sphere. Clearly illustrate each step of the derivation.
5. Define torque and explain how does it differ from force in linear motion?

Numericals:

1. A car mechanic applies a force of 800 N to a wrench for the purpose of loosening a bolt. He applies the force which is perpendicular to the arm of the wrench. The distance from the bolt to the mechanic's hand is 0.40 m. Find out the magnitude of the torque applied? **(320 N .m)**
2. A car accelerates uniformly from rest and reaches a speed of 22 m/s in 9 s. If the diameter of a tire is 58 cm, find
   (a) the number of revolutions the tire makes during this motion, assuming no slipping, and
   (b) the final rotational speed of the tire in revolutions per second. **(55 rev, 12 rev /s)**

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. Work done by centripetal force is always:
   (a) Maximum (b) Minimum
   (c) Zero (d) None of these
2. A body of mass 5 kg is moving with a momentum of 10 kg m/s . A force of 0.2 N acts on it in the direction of motion of the body for 10 seconds. The increase in kinetic energy is:
   (a) 2.8 J (b) 3.2 J (c) 3.8 J (d) 4.4 J
3. The kinetic energy of a light and a heavy object is same. Which object has maximum momentum?
   (a) Light object (b) Heavy object
   (c) Both have same momentum (d) N.O.T
4. Two bodies of mass 1 kg and 2 kg have equal momentum. Then the ratio of their kinetic energies is:
   (a) 2 :1 (b) 3:1 (c) 1 :3 (d) 1:1
5. A body fallen from height h . After it has fallen a height h/2 , its will possess :
   (a) only potential energy (b) only kinetic energy
   (c) half potential half kinetic energy (d) more kinetic and less potential
6. Which of the following quantity can be calculated by multiplying force and velocity ?
   (a) acceleration (b) power (c) torque (d) work
7. The minimum velocity given to an object so that it emerges out from the gravitational field of earth is about _____ :
   (a) 11.2 km/s (b) 15.3 km/s (c) 5 km/s (d) 9.8 km/s
8. When one joule of work is done on a body in one second, power of body is said to be :
   (a) One watt (b) 0.5 watt (c) zero (d) 100 watt
9. The absolute potential energy of an object depends on:
   (a) The object's mass and height (b) The object's mass and speed
   (c) The object's shape and size (d) The object's color and temperature
10. The escape velocity of a planet depends on which of the following factors?
    (a) The mass of the planet only
    (b) The radius of the planet only
    (c) Both the mass and the radius of the planet
    (d) The density of the planet

Section (B): Structured Questions

CRQs:

1. How does work relate to the transfer of energy?
2. How is power related to work and time?
3. What is the difference between average power and instantaneous power?



Section (A): Multiple Choice Questions (MCQs)

1. A completely submerged object always displaces its own_____
   a) weight of fluid
   b) volume of a fluid
   c) density of a fluid
   d) Area of fluid
2. The pressure exerted on the ground by a man is greatest when:
   a) he stands with both feet flat on the ground
   b) he stands flat on one foot
   c) he stands on the toes of one foot
   d) he lies down on the ground
3. In a stationary homogeneous liquid:
   a) pressure is the same at all points
   b) pressure depends on the direction
   c) pressure is independent of any atmospheric pressure on the upper surface of the liquid
   d) pressure is the same at all points at the same level
4. One piston in a hydraulic lift has an area that is twice the area of the other. When the pressure at the smaller piston is increased by Δp the pressure at the larger piston:
   a) increases by $2\Delta p$
   b) increases by $\Delta p/2$
   c) increases by Δp
   d) increases by $4\Delta p$
5. In a vacuum, an object has
   a) No buoyant force
   b) no mass
   c) no weight
   d) none of these
6. The pressure at the bottom of a pond does NOT depend on
   a) Water density
   b) the depth of the pond
   c) the surface area of the pond
   d) None of these
7. A rock suspended by a weighing scale weighs 5N out of water and 3 N when submerged in water. What is the buoyant force on the rock?
   a) 3 N
   b) 5 N
   c) 8N
   d) 15 N
8. "An object completely submerged in a fluid displaces its own volume of fluid". This is:
   a) Pascal's paradox
   b) Archimedes' principle
   c) Pascal's principle
   d) true, but none of the above
9. Salt water has greater density than freshwater. A boat floats in both freshwater and saltwater. The buoyant force on the boat in salt water is ---------- that in freshwater.
   a) equal to
   b) smaller than
   c) larger than
   d) same as
10. You fill a tall glass with ice and then add water to the level of the glass's rim, so some fraction of the ice floats above rim. When the ice melts, what's happens to the water level?
    a) the water overflows the rim
    b) the water level drops below the rim
    c) the water level stays at the rim
    d) it depends on the difference in density between water and ice



Section (A): Multiple Choice Questions (MCQs)

1. For an incompressible fluid, the flow rate is
   a) equal for all surfaces.
   b) constant throughout the pipe.
   c) greater for the larger parts of the pipe.
   d) none of the above

2. Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sum of the pressure and energy of motion per unit volume is
   a) increasing with time
   b) decreasing with time
   c) constant
   d) varying with time

3. Which of the following is associated with the law of conservation of energy in fluids?
   a) Archimedes' principle
   b) Bernoulli's principle
   c) Pascal's principle
   d) equation of continuity

4. As the speed of a moving fluid increases, the pressure in the fluid
   a) increases
   b) remains constant
   c) decreases
   d) may increase or decrease, depending on the viscosity

5. If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?
   a) Increases
   b) Decreases
   c) Remains the same
   d) Depends on the fluid density

6. A sky diver falls through the air at terminal velocity. The force of air resistance on him is
   a) half his weight
   b) equal to his weight
   c) twice his weight
   d) Cannot be determined from the information given.

7. Wind speeding up as it blows over the top of a hill
   a) Increases atmospheric pressure there.
   b) decreases atmospheric pressure there.
   c) doesn't affect atmospheric pressure there.
   d) equal's atmospheric pressure.

8. A fluid is undergoing "incompressible" flow. This means that:
   a) the pressure at a given point cannot change with time
   b) the velocity at a given point cannot change with time
   c) the velocity must be the same everywhere
   d) the pressure must be the same everywhere
   e) the density cannot change with time or location

9. A fluid is undergoing steady flow. Therefore:
   a) the velocity of any given molecule of fluid does not change
   b) the pressure does not vary from point to point
   c) the velocity at any given point does not vary with time
   d) the density does not vary from point to point

10. The equation of continuity for fluid flow can be derived from the conservation of:
    a) energy
    b) mass
    c) volume
    d) pressure

Section (B): Structured Questions

CRQ's:

1. What is difference between streamline and turbulent flow?
2. Would a drinking straw work in space where there is no gravity? Explain.
3. Why do airplanes take off into wind?
4. Describe terminal velocity in liquids.
5. Discuss the significance of Reynolds number.
6. State Bernoulli's principle.
7. Give two applications of Bernoulli's principle.
8. 'Fluid flow is turbulent rather than laminar', support this statement.
9. Discuss importance of Stokes law.
10. Justify spin of ball in Bernoulli's principle.

ERQ's:

1. Derive equation of continuity. Also show its physical significance.
2. Derive Bernoulli's equation.
3. Discuss viscous force in fluids.
4. Define fluid dynamics and explain its significance in the study of fluids. How does it differ from fluid statics?
5. Discuss the concept of Reynolds number and its significance in fluid dynamics. Explain how Reynolds number relates to the transition between laminar and turbulent flow.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. A 2μC point charge is located a distance "d" away from 6μC point charge, what is the ratio of F_{12}/F_{21}?
   (a) 1/3 (b) 3
   (c) 1 (d) 12
2. The minimum charge on an object cannot be less than:
   (a) 1.6×10^{-19}C (b) 3.2×10^{-19}C
   (c) 9.1×10^{9} (d) No definite value exist
3. Two charges are placed at a certain distance. If the magnitude of each charge is doubled the force will become
   (a) 1/4th of its original value (b) 4 times of its original value
   (c) 1/8th of its original value (d) 8 times of its original value
4. Which of the following can be deflected while moving in the electric field?
   (a) neutron (b) photon
   (c) electron (d) (a) and (b)
5. The flux through a flat surface of are "A" in a uniform electric field "E" is maximum when the surface area is:
   (a) Parallel to E (b) perpendicular to E
   (c) placed 45^0 to E (d) placed 60^0 to E
6. The product of charge "q" and small separation "d" between two charges of same magnitude and opposite in nature is known as:
   (a) Electric dipole (b) Moment arm
   (c) Electric dipole moment (d) Flux of electric field
7. 12 J of work is to be done against an existence electric field to take a charge of 0.01 C from one-point A to another point B. The potential difference between B and A is
   (a) 120 V (b) 1200 V
   (c) 1.2 V (d) 12 V
8. The force between two charges placed in air is F. if air is replaced by a medium of relative permittivity ϵ_r then force is reduced to:
   (a) $F\epsilon_r$ (b) F/ϵ_r
   (c) ϵ_r/F (d) $\epsilon\epsilon_r$
9. The negative gradient of the potential is:
   (a) potential energy (b) voltage
   (c) electric field intensity (d) electric flux
10. The electric flux through a plane area will be half of its maximum value when area is held at angle of _____ with electric field
    (a) 30° (b) 60°
    (c) 45° (d) 90°

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. The capacitance of a capacitor is NOT influenced by:
   (a) Plate thickness (b) Plate area
   (c) Plate separation (d) Nature of the dielectric
2. What is the value of capacitance of a capacitor which has a voltage of 4V and has 16C of charge?
   (a) 2F (b) 4F (c) 6F (d) 8F
3. Capacitors are used in electric power supply system to:
   (a) Improve power factor (b) Reduce line current
   (c) Provide voltage stability (d) switching
4. In a variable capacitor, capacitance can be varied by:
   (a) Turning the rotatable plates in or out (b) Changing the plates
   (c) Sliding the rotatable plates (d) Changing the material of plates
5. Energy stored in the capacitor is:
   (a) E = ¼ CV (b) E = ½ CV^2 (c) E = CV^2 (d) E = ½ CV
6. The time constant of a series RC circuit consisting of $100\mu F$ capacitor in series with a 100Ω resistor is:
   (a) 0.1 s (b) 0.1 ms (c) 0.01 s (d) 0.01 ms
7. The charging of a capacitor through a resistance follows:
   (a) linear law (b) square law
   (c) exponential law (d) none of the above
8. When the total charge in a capacitor is doubled, the energy stored
   (a) remains the same (b) is doubled
   (c) is halved (d) is quadrupled
9. The capacitance C is charged through a resistor R. The time constant of the charging circuit is given by:
   (a) C/R (b) 1/RC (c) RC (d) R/C
10. Capacitor blocks:
    (a) alternating current (b) direct current
    (c) both alternating and direct current (d) neither alternating nor direct current

Section (B): Structured Questions

CRQs:

1. State the factors on which the capacitance of a parallel plate capacitor depends.
2. Explain what is meant by dielectric constant (relative permittivity). State two physical properties desirable in a material to be used as the dielectric in acapacitor.
3. Drive expressions for the combined capacitance of two capacitors (a) connected in series, (b) connected in parallel.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. Kirchhoff's laws are useful in determining:
   a) current flowing in a circuit b) emf and voltage drops in a circuit
   c) power in a circuit d) only emf in a circuit
2. The resistance of a superconductor is:
   a) finite b) infinite
   c) changes with every conductor d) zero
3. Reciprocal of resistance is called:
   a) conductance b) resistivity c) resonance d) capacitance
4. The graphical representation of Ohms law is:
   a) parabola b) hyperbola c) ellipse d) straight line
5. A potential difference is applied across the ends of a wire. If the potential difference is doubled, then the drift velocity of free electrons will:
   a) be quadrupled b) be doubled c) be halved d) remain unchanged
6. Internal resistance is the resistance offered by:
   a) Capacitor b) resistor c) Conductor d) source of emf
7. Power dissipation in a resistor can be calculated using which formula:
   a) $P = V^2 / R$ b) $P = I^2 \times R$ c) $P = V \times I$ d) $P = R / (V \times I)$
8. What is a potentiometer primarily used for?
   a) Measuring electric current
   b) Measuring electric charge
   c) Measuring potential difference (voltage)
   d) Measuring electric resistance
9. A heat-sensitive device whose resistivity changes with the change in temperature is called:
   a) conductor b) resistor c) thermistor d) thermometer
10. A wire of uniform area of cross-section A length L and resistance R is cut into two parts. The resistivity of each part:
    a) Becomes zero b) is halved c) Is doubled d) remains same

CRQs:

1. Why is the terminal voltage of a cell less than its emf?
2. Why is a potentiometer preferred over a voltmeter for determining the emf of a cell?
3. Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more? Justify your answer.
4. Explain why the terminal potential of a battery decreases when the current drawn from it in increased?
5. What are thermistors? Write their importance.
6. State Kirchhoff's Laws.
7. If Copper and Aluminum wires of the same length have same resistance, which has the larger diameter? And why?

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. Two simple pendulums A and B with same lengths, and equal amplitude of vibrations, but the mass of A is twice the mass of B, their period are T_A and T_B and energies are E_A and E_B respectively. Choose the correct statement.
   a) $T_A = T_B$ and $E_A > E_B$ b) $T_A < T_B$ and $E_A > E_B$
   c) $T_A > T_B$ and $E_A < E_B$ d) $T_A = T_B$ and $E_A < E_B$
2. In order to double the period of a simple pendulum:
   a) Its length should be doubled b) Its length should be quadrupled
   c) The mass should be doubled d) The mass should be quadrupled
3. A simple harmonic oscillator has amplitude A and time period t. Its maximum speed is:
   a) $\frac{4A}{t}$ b) $\frac{2A}{t}$ c) $\frac{4\pi A}{t}$ d) $\frac{2\pi A}{t}$
4. A spring attached by a load of weight W is vibrating with a period T. If the spring is divided in four equal parts and the same load is suspended from one of these parts, the new time period is:
   a) $\frac{T}{4}$ b) 2T c) $\frac{T}{2}$ d) 4T
5. The total energy of a particle executing simple harmonic motion is proportional to:
   a) frequency of oscillation b) maximum velocity of motion
   c) amplitude of motion d) square of amplitude of motion
6. A child swinging on a swing in sitting position, stands up, then the time period of the swing will:
   a) Increase b) decrease c) remains the same
   d) increases if the child is long and decreases if the child is short
7. If a body oscillates at the angular frequency ω_d of the driving force, then the oscillations are called:
   a) Forced oscillations b) Coupled oscillations
   c) Free oscillations d) Maintained oscillations
8. A simple harmonic oscillator with a natural frequency ω_N is force to oscillate with a driving frequency ω_d. The Resonance occurred when:
   a) $\omega_N < \omega_d$ b) $\omega_N > \omega_d$ c) $\omega_N = \omega_d$ d) $\omega_N \approx \omega_d$
9. In vehicles, shock absorber reduced the jerks:
   a) The shock absorber is the application of damped oscillations.
   b) Damping effect is due to the fractional loss of energy
   c) Shock absorbers in vehicles reduced jerk
   d) All of these
10. A heavily damped system has a fairly flat resonance curve in:
    a) An acceleration time graph b) An amplitude frequency graph
    c) Velocity time graph d) Distance-time graph

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. The speed v of a wave represented by $y = A\sin(\omega t - kx)$ is:
   a)k/ω b) ω/k c) ωk d) 1/ωk
2. Two sound waves are $y = A\sin(\omega t - kx)$ and $y = A\cos(\omega t - kx)$. The phase difference between the two waves is:
   a) π/2 b) π/4 c) π d) 0
3. If v_a, v_h and v_m are the speeds of sound in air, hydrogen gas, and a metal at the same temperature, then:
   a) $v_a > v_h > v_m$ b) $v_m > v_h > v_a$ c) $v_h > v_m > v_a$ d) $v_h > v_a > v_m$
4. The speed of sound in air at STP is 332 m/s. If the air pressure becomes double at the same temperatue, the speed of sound become:
   a) 1382 m/s b) 664m/s c) 332 m/s d) 166 m/s
5. How does the speed of sound v in air depend on the atmospheric pressure P:
   a) $v \propto P^0$ b) $v \propto P^{-1}$ c) $v \propto P^2$ d) $v \propto P^1$
6. The speed of sound in a gas is proportional to:
   a) square root of isothermal elasticity b) square root of adiabatic elasticity
   c)isothermal elasticity d)adiabatic elasticity
7. The length of a pipe closed at one end is L. In the standing wave whose frequency is 7 times the fundamental frequency, what is the closest distance between nodes?
   a) $\frac{1}{14}$ L b) $\frac{1}{7}$ L c) $\frac{2}{7}$ L d) $\frac{4}{7}$ L
8. A 620 Hz frequency song of a ice cream trolley approaches with speed v to a boy standing at the door of his house is heard with frequency f_1. If the trolley is stopped and the boy approaches to the ice cream trolley with same speed v; the boy now hears the sound with frequency f_2. choose the correct statement:
   a) $f_1 = f_2$; both are greater than 620Hz b) $f_2 > f_1 > 620$Hz
   c) $f_1 = f_2$; both are lesser than 620Hz d) $f_1 > f_2 > 620$Hz.
9. The speed of sound in a gas in which two waves of wavelength 50 cm and 50.4 cm produce 6 beats per second is:
   a) 338 m/s b) 350 m/s c) 378 m/s d) 400 m/s
10. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 minutes, then its wave length is:
    a) 13.85 m b) 25.3 m c) 41.5 m d) 57.2 m

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. If the wavelength of an electromagnetic wave is about the diameter of a cricket ball, what type of radiation is it.
   a) X-ray b) Ultraviolet
   c) Radio waves d) Visible light
2. Electromagnetic waves from an unknown source in space are found to be diffracted when passing through gaps of the order of 10^{-5} m, which type of the wave are they most likely to be?
   a) microwaves b) Ultraviolet
   c) Radio waves d) infra-red waves
3. Huygens's conception of secondary waves
   a)helps us to find the focal length of a thick lens
   b) is a geometrical method to find a wavefront
   c)is used to determine the velocity of light
   d) is used to explain the polarization of light
4. Interference fringes are produced using monochromatic light of same intensity from a double slit screen. If the intensity of light emerging from one of the slit is reduced, the effect on interference pattern will be
   a)All the dark and bright fringes become brighter.
   b) All the dark and bright fringes become darker.
   c)Bright fringes become brighter and dark fringes become darker.
   d) Bright fringes become darker and dark fringes become brighter.
5. In Young's experiment when the distance between slits and screen is doubled, while separation of slits is halved, then fringe width will be;
   a) 4 times b) ¼ times c) doubled d) unchanged
6. A ray of light passes from air into water. Striking the surface of the water with an angle of incidence 45°. Which of these quantities change as the light enters the water.i) Wavelength ii) frequency iii) speed of propagation iv) direction of propagation
   a) i and ii only. b) iii and iv only.
   c) i, iii, and iv only. d) all of them.
7. A hill separates a television (TV) transmitter from a house. The Transmitter cannot be seen from the house but still the TV in the house has good reception. What wave phenomena make it possible?
   a) Coherence of waves b) Diffraction of waves
   c) Interference of waves d) Refraction of waves
8. Monochromatic light is incident on a diffraction grating and a diffraction pattern is observed. Which effects is observed by replacing the grating with one that has more lines per millimeter?
   a)Number of maxima decreases with decrease in angle between first and second order maxima.

b) Number of maxima decreases with increase in angle between first and second order maxima.
c)Number of maxima increases with decrease in angle between first and second order maxima.
d) Number of maxima increases with increase in angle between first and second order maxima.

9. Optically active substances are those substances which
a)produce polarized light
b) rotate the plane of polarization of polarized light
c)produce double refraction
d) convert a plane polarized light into circularly polarized light

10. Plane polarized light is passed through a Polaroid. On viewing through the Polaroid we find that when Polaroid is given one complete rotation about the direction of light
a)The intensity of light gradually decreases to zero and remains at zero.
b) The intensity of light gradually increases to maximum and remains at maximum.
c)There is no change in the intensity of light.
d) The intensity of light varies such that it is twice maximum and twice zero.

CRQs

1. In every day experience, visible light seem to travel in straight lines while radio waves do not, Explain.
2. Explain why two waves of significantly different frequencies cannot be coherent?
3. In a Young's double slit experiment, how the interference phenomenon is affected by changing the slits separation and the distance between the slits and screen?
4. Explain how the double-slit experiment provides evidence for the wave nature of light.
5. Discuss the concept of monochromatic light in the context of Newton's rings.
6. Explain why monochromatic light is preferred for obtaining clear and well-defined interference patterns in the experiment.
7. Discuss how the interference of light waves leads to the formation of bright and dark rings in the experiment.

ERQs

1. Explain the concept of interference in physical optics. Discuss constructive and destructive interference.
2. Discuss the conditions required for interference and provide examples of interference in daily life.
3. Describe the setup and working principle of the Michelson interferometer. Also explain how the Michelson interferometer can be used to measure the wavelength of monochromatic light.
4. Describe the setup and procedure of the diffraction of X-rays through a crystal experiment.

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. In Radio and Television broadcast, the information signal is in the form of :
   a) analog signal b) digital signal
   c) Both analog & digital signals d) neither analog nor digital signal
2. A communication channel consists of:
   a) transmission line only b) optical fibre only
   c) free space only d) All of them
3. Voltage signal generated by a microphone is:
   a) digital in nature b)analog in nature
   c) hybrid in nature d) consists of bits & bytes
4. As compared to sound waves frequency of radio waves is:
   a) higher b) equal c) lower d) may be higher or lower
5. The process of superimposing signal frequency on the carrier wave is known as:
   a) transmission b)detection c)reception d)modulation
6. What is the frequency range of signals that can be transmitted in case of twisted pair of wires?
   a)10MHz. to 15MHz. b)5MHz. to 10MHz.
   c)100Hz. to 5MHz. d)10Hz. to 100Hz.
7. The maintenance of a satellite's orbital position is called:
   a) maintenance keeping b) station keeping
   c)station maintenance d)attitude control
8. Process of mapping the sampled analog voltage values to discrete voltage levels is called:
   a) sampling b)sampling frequency
   c) quantizing d)encoding
9. AM is used for broadcasting because:
   a) it requires less transmitting power compare with other systems
   b) it is more noise immune than other modulation system
   c)No other modulation can provide the necessary bandwidth faithful transmission
   d) its use avoids receiver complexity
10. Data in compact disc is stored in the form of :
    a) analog signal b)digital signal
    c) noise d)both (a) &(b)